

and themes.

`/boot/grub2/grub.cfg`

This has the main GRUB2 configuration file. This file is auto generated, and should normally not be edited manually. As a convenience, `/etc/grub2.cfg` is a symbolic link to this file.

`/etc/grub.d/`

This directory contains a helper script to generate a main GRUB2 configuration file.

`/etc/defaults/grub`

This file contains variables used in the generation of the main GRUB2 configuration file, as shown in Figure 2.

`/boot/grub2/grubenv`

This contains a file of exactly 1KiB, which is used as storage for variables, such as a default or 'saved' boot entry.

The `grub2-editenv` tool can be used to read and modify these settings.

The Master Boot Record (MBR)

In order to boot GRUB2 from the disk on a BIOS system, there are two options—store the first part of the GRUB2 boot loader in the Master Boot Record (MBR) of a disk, or store it in the first sector of a partition that is marked as 'bootable'.

The problem with the MBR

There is a major constraint with using the MBR. A typical MBR is only 512 bytes in size, and part of that space is used for the partition table of that disk, leaving only 446 bytes for the boot loader.

To work around this issue, GRUB2 can use the 'boot track', 'MBR gap', or 'embedding area' on the disk to store the extra modules and files it needs. This is the space between the MBR and the first partition on the disk. In order to work reliably, there needs to be at least 31KiB of space available this way (63 sectors on a 512-byte sector disk). If a disk has been partitioned by Anaconda, the first partition will usually start at sector 2048, leaving roughly 1MiB of space for the embedding area, which is ample room for GRUB2.



Note: The `ibi` (or binary prefix) units KiB, MiB, GiB are based on 2^{10} (1024) whereas standard decimal units (KB, MB, GB) are based on 10^3 (1000). In this article, we have used the units KiB, MiB and TiB.
 1KB= 1000 bytes 1KiB=1024 bytes
 1MB=1000KB 1MiB=1024KB
 1GB=1000MB 1GiB=1024MB
 1TB=1000GB 1TiB= 024GB

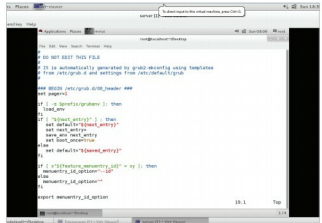


Figure 1: The configuration file of boot loader



Figure 2: Variables used in the generation of the main GRUB2 configuration file

Configuring GRUB2

Under normal circumstances, an administrator should not have to manually configure the GRUB2 boot loader. When a new kernel is installed, it will be added to the configuration automatically and when a kernel is removed, the corresponding entry in the boot loader menu is removed automatically as well.

An administrator might want to tweak some parameters that are passed into the kernel at startup by GRUB2. The best way to do this is by editing `/etc/defaults/grub` and then forcing a recreation of the main GRUB2 configuration file.

The settings in `/etc/defaults/grub` that are of interest are listed here.

GRUB_TIMEOUT

This is the number of seconds the GRUB2 menu is displayed before the default entry is booted automatically.

GRUB_DEFAULT

This is the number of the default entry that should be started whenever a user does not select another entry, which is